

FinalReport

Introduction

Variable selection is a tool widely used when modelling to find parsimonious and predictive models. Many methods are available for variable selection with different use cases, strengths, and weaknesses. Because variable selection may sometimes be of use, for example for predictive modelling, it is of interest to investigate how these methods compare with varying correlation structures and sample sizes. For this project, we will compare backwards selection using p-values, AIC, and BIC, as well as regression using lasso and elastic net, using simulated datasets with known correlations between the variables, and performance metrics including true and false positives, type I and II error rate of variables selected (using $\alpha=.05$), confidence interval coverage, and bias. We want to test whether these methods perform similarly or whether the error rate, bias, and coverage differ across the models, with 2 sample size cases, and 3 correlation structure cases.

Methods

Describe the simulation flow

We are comparing the 7 methods for variable selection. Backwards selection using p-values with $\alpha = .05$ removes variables one model fit at a time, removing the term with the highest p-value until all included terms have $p < .1$. Backwards selection using AIC or BIC perform similarly, stopping once a model's AIC or BIC is not improved (decreased) by the exclusion of additional variable. We also implemented LASSO (Least Absolute Shrinkage and Selection Operator) regression which shrinks unnecessary or non-predictive coefficients towards zero. We considered two penalty term options, either the λ_{min} such that λ_{min} minimizes the mean cross-validated error, and the λ_1 value 1 standard deviation larger than λ_{min} . These two choices are conventionally used, since the 1 standard error case provides a parsimonious model with error no more than 1 standard error that of λ_{min} model, since minimizing the error may lead to overfitting (Krstajic 2014). We used these same λ options for elastic-net, which we implemented using $\alpha = .5$, because we wanted to implement elastic net as distinctively from ridge regression ($\alpha = 0$) and LASSO ($\alpha = 1$) as possible. The p-value method was implemented using the `ols_step_backward_p()` function in the `olsrr` package in R, AIC and BIC using the `step()` function with $k=2$ and $k = \log(n)$ respectively, and the LASSO and E-net methods were implemented using the above described parameters.

```
methods_matrix <- matrix(c(
  "P-values", "ols_step_backward_p()", "olsrr", "p=.1",
  "AIC", "step()", "", "k=2",
  "BIC", "step()", "", "k = log(n)",
  "LASSO", "glmnet, cv.glmnet", "glmnet", "lambda = lambda.min, alpha = 1",
  "LASSO", "glmnet, cv.glmnet", "glmnet", "lambda = lambda.1se, alpha = 1",
  "E-net", "glmnet, cv.glmnet", "glmnet", "lambda = lambda.min, alpha = .5",
  "Enet", "glmnet, cv.glmnet", "glmnet", "lambda = lambda.1.se, alpha = .5"), ncol = 4, byrow = TRUE)
colnames(methods_matrix) <- c("Method", "Functions", "Package", "Parameters")
kable(methods_matrix, caption = "Summary of methods compared")
```

Table 1: Summary of methods compared

Method	Functions	Package	Parameters
P-values	ols_step_backward_p()	olsrr	p=.1

Method	Functions	Package	Parameters
AIC	step()		k=2
BIC	step()		k = log(n)
LASSO	glmnet, cv.glmnet	glmnet	lambda = lambda.min, alpha = 1
LASSO	glmnet, cv.glmnet	glmnet	lambda = lambda.1se, alpha = 1
E-net	glmnet, cv.glmnet	glmnet	lambda = lambda.min, alpha = .5
Enet	glmnet, cv.glmnet	glmnet	lambda = lambda.1se, alpha = .5

Because we were interested in both coverage and bias in our simulations, we used two approaches from Morris 2019 to calculate needed n_{sim} based on these two measures. Attempting these two methods provided divergent results, so we chose to use $n_{sim} = 10000$ to ensure adequate power, as this benchmark is often used and recommended (Morris 2019). For each iteration, we generated 6 datasets using the hdrn package in R, with sample sizes 250 and 500, as well as exchangeable correlation structure within the design matrix with $\rho = 0$ (no correlation), $\rho = .35$, and $\rho = .7$. For each dataset, 5 of the 20 variables in the design matrix had a true association with the outcome vector y , with true coefficients for variables V_1, V_2, V_3, V_4, V_5 as $1/6, 1/3, 1/2, 2/3$, and $5/6$ respectively. With the exchangeable correlation structure, V_1 through V_{20} are assumed to have a between-variable correlation of ρ , and sampled accordingly. We then performed variable selection for each of the 6 cases using the 7 described methods. Multiple linear regression models were refit in R for y regressed on the selected variables from each method. From these refit models, we retained information on the bias of the coefficients vs. their true values, the amount of times that the models included V_1 through V_5 , the variables with true associations, the amount of times the models included V_6 through V_{20} (the null variable group), and how often type 1 and type 2 errors were committed. We calculated type 1 error rates two ways. The first way was at the model level, e.g., how often was at least one variable in the null group included in the regression model with a significant p-value (using $\alpha = .5$ as a threshold since this is often used in medical studies) and also as the average number of type 1 errors committed by a modelling approach for a given case. Similarly, we investigated type 2 error at the model level and reported the average number of type 2 errors committed by modelling approach and case. Reporting the number of errors per model allows us to compare approaches at a more granular level since, for example, there is a difference between a model with 3 insignificant variables included and a model with 1, though both would be treated the same at the model level. We also investigated type 2 error rates for the individual variables since we had a range of coefficient strengths across V_1 through V_5 . In other words, the TPR and FPR were reported at the model level, and type 1 and type 2 error introduced more granularity as to which variables were significant, not only whether models kept all truly significant variables significant in the model and vice versa. Reporting these measures this way allows us to understand model patterns more robustly since we can look at trends at both the model level and variable level. Both type 1 and type 2 error rates are handled as though no model selection had been done, which will inflate type 1 error rates and underestimate standard error (Shen 2024). Since this naive approach is often used in practice, it is beneficial to see how methods compare in this context. Simulation procedure was parallelized using doRNG package in R, and we used seed 23192319 to ensure reproducible results. Code for simulating the data and producing tables and plots is available at <https://github.com/jkebe2002/BIOS6624/tree/9c435c37471509d45ecdb37552575dbf7bd10cd7/P4>.

How is the data generated What metrics are you interested in How were they calculated Reproducibility Choices for lambda, alpha, n_{sim} , etc.

Results

Comment on coverage, bias of different variables coverage bias for different cases / methods

MENTION BIAS OF THE NULL VARIABLES!

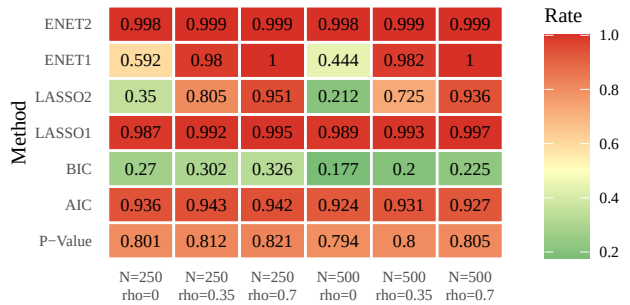
FP,TP error rates Secondary table of error rates by variables

```
library(patchwork)
(p_trueP | p_falseP) / (p_type1_m | p_type2_m) / (p_type1 | p_type2)
```

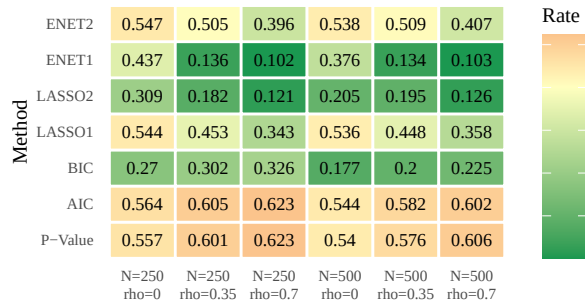
True Positive Rate:
All significant variables included in Model



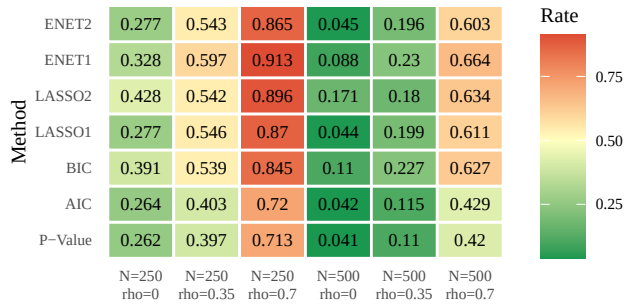
False Positive Rate:
Any non-significant variables included



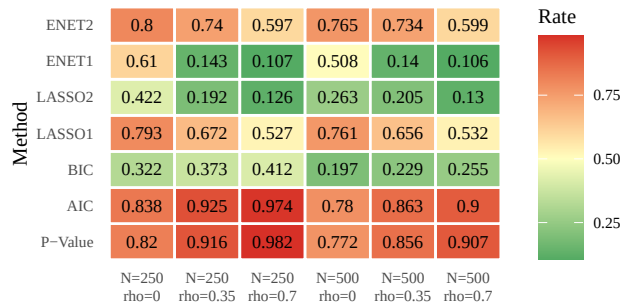
Type 1 Error (Model Level)



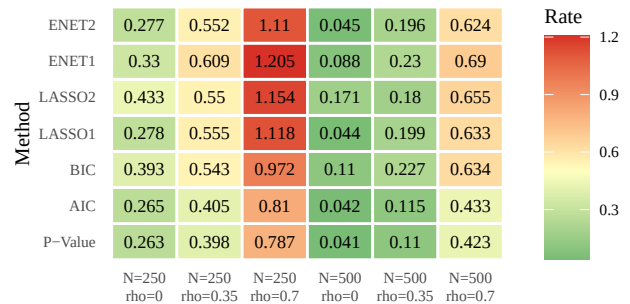
Type 2 Error (Model Level)

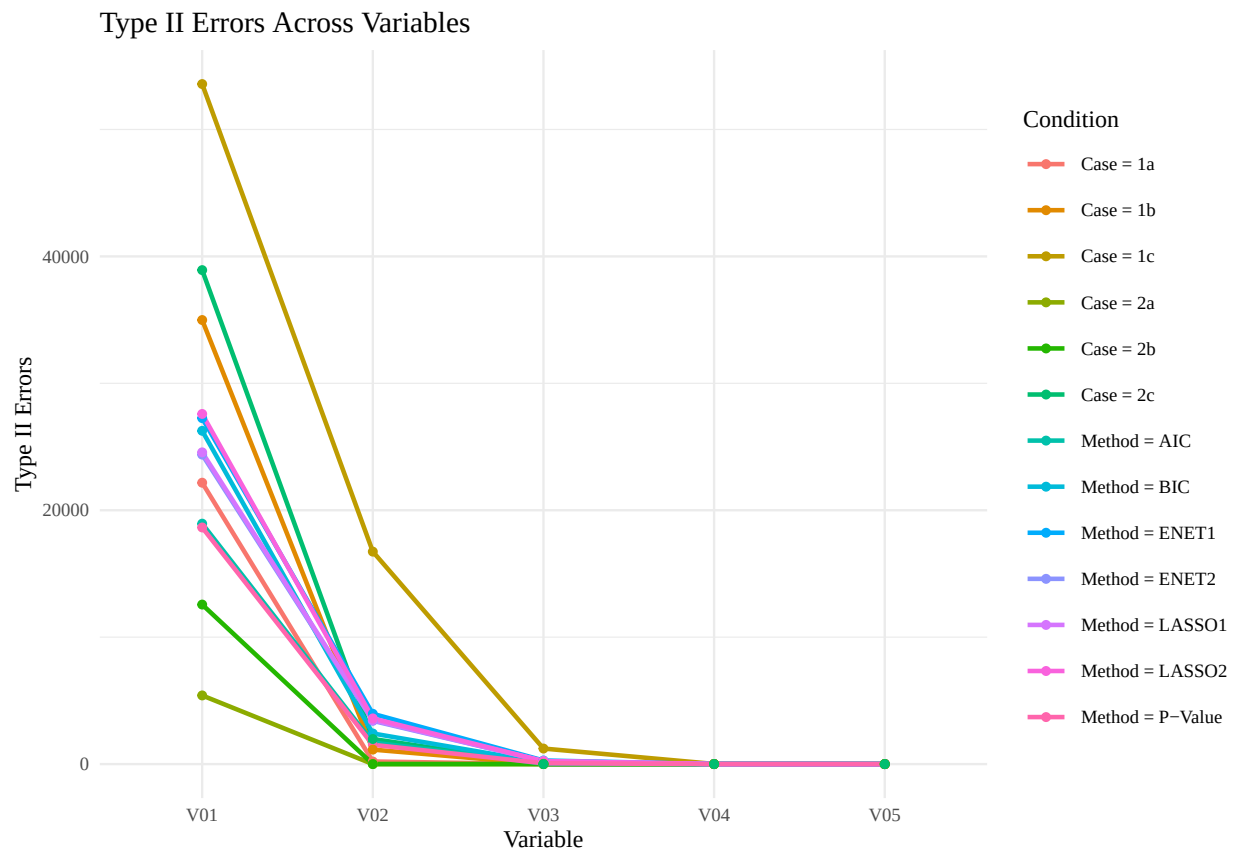


Type 1 Error (# Errors per Model)



Type 2 Error (# Errors per Model)





Plots?

Table 2: Bias and Coverage by Method, Variable, and Correlation Structure

N	Method	V01		V02		V03		V04		V05	
		Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.
rho = 0											
N = 250											
N=250	P-Value	-0.012	0.800	-0.001	0.944	-0.001	0.949	0.000	0.941	0.001	0.943
N=250	AIC	-0.007	0.854	-0.001	0.943	-0.001	0.947	0.000	0.939	0.001	0.942
N=250	BIC	-0.041	0.585	-0.001	0.946	-0.001	0.951	-0.001	0.944	0.001	0.944
N=250	LASSO1	-0.004	0.908	-0.003	0.943	-0.002	0.946	-0.002	0.939	-0.001	0.940
N=250	LASSO2	-0.047	0.562	-0.004	0.945	-0.003	0.951	-0.002	0.943	-0.001	0.944
N=250	ENET1	-0.028	0.694	-0.004	0.943	-0.003	0.949	-0.003	0.942	-0.002	0.945
N=250	ENET2	-0.002	0.929	-0.003	0.943	-0.002	0.947	-0.002	0.940	-0.001	0.941
N = 500											
N=500	P-Value	-0.001	0.948	-0.001	0.947	0.000	0.949	0.000	0.946	0.000	0.948
N=500	AIC	-0.001	0.948	-0.001	0.946	0.000	0.949	0.000	0.947	0.000	0.947
N=500	BIC	-0.010	0.865	-0.001	0.947	0.000	0.952	0.000	0.948	0.000	0.950
N=500	LASSO1	-0.001	0.948	-0.001	0.946	0.000	0.950	-0.001	0.946	0.000	0.946
N=500	LASSO2	-0.018	0.805	-0.001	0.948	0.000	0.951	0.000	0.948	0.000	0.949
N=500	ENET1	-0.008	0.890	-0.001	0.947	0.000	0.951	-0.001	0.946	0.000	0.950
N=500	ENET2	-0.001	0.948	-0.001	0.946	0.000	0.950	-0.001	0.947	0.000	0.947
rho = 0.35											
N = 250											
N=250	P-Value	-0.022	0.679	0.002	0.941	0.002	0.936	0.002	0.934	0.003	0.939
N=250	AIC	-0.013	0.751	0.002	0.941	0.001	0.935	0.001	0.935	0.002	0.939
N=250	BIC	-0.058	0.444	0.006	0.938	0.008	0.935	0.008	0.933	0.008	0.938
N=250	LASSO1	-0.023	0.923	-0.023	0.935	-0.024	0.932	-0.024	0.930	-0.024	0.932
N=250	LASSO2	-0.034	0.795	-0.024	0.934	-0.025	0.931	-0.025	0.927	-0.025	0.932
N=250	ENET1	-0.034	0.907	-0.034	0.931	-0.035	0.929	-0.035	0.927	-0.035	0.929
N=250	ENET2	-0.021	0.932	-0.022	0.937	-0.022	0.934	-0.023	0.933	-0.022	0.933
N = 500											
N=500	P-Value	-0.003	0.904	0.000	0.942	0.000	0.942	0.000	0.942	0.000	0.939
N=500	AIC	-0.001	0.928	0.000	0.942	0.000	0.943	0.000	0.943	0.000	0.940
N=500	BIC	-0.022	0.745	0.003	0.942	0.003	0.943	0.003	0.944	0.003	0.940
N=500	LASSO1	-0.016	0.934	-0.017	0.932	-0.017	0.935	-0.017	0.934	-0.017	0.933
N=500	LASSO2	-0.017	0.926	-0.016	0.935	-0.016	0.937	-0.016	0.935	-0.016	0.931
N=500	ENET1	-0.023	0.930	-0.025	0.928	-0.025	0.930	-0.024	0.928	-0.025	0.926
N=500	ENET2	-0.014	0.936	-0.016	0.936	-0.016	0.938	-0.015	0.937	-0.016	0.936
rho = 0.7											
N = 250											
N=250	P-Value	-0.047	0.415	-0.007	0.872	0.005	0.935	0.006	0.933	0.006	0.931
N=250	AIC	-0.031	0.512	-0.003	0.904	0.003	0.937	0.003	0.932	0.003	0.935
N=250	BIC	-0.092	0.201	-0.022	0.755	0.018	0.923	0.022	0.920	0.022	0.921
N=250	LASSO1	-0.040	0.811	-0.048	0.920	-0.049	0.924	-0.048	0.925	-0.048	0.923
N=250	LASSO2	-0.052	0.704	-0.058	0.911	-0.058	0.916	-0.057	0.918	-0.057	0.917
N=250	ENET1	-0.051	0.881	-0.059	0.920	-0.060	0.926	-0.059	0.927	-0.058	0.927
N=250	ENET2	-0.037	0.841	-0.045	0.922	-0.046	0.926	-0.045	0.927	-0.045	0.926
N = 500											
N=500	P-Value	-0.025	0.653	0.001	0.938	0.003	0.940	0.003	0.936	0.003	0.933
N=500	AIC	-0.016	0.728	0.001	0.937	0.001	0.941	0.002	0.936	0.002	0.934
N=500	BIC	-0.072	0.361	0.008	0.924	0.013	0.931	0.014	0.929	0.013	0.925
N=500	LASSO1	-0.031	0.918	-0.033	0.925	-0.033	0.930	-0.032	0.921	-0.032	0.921
N=500	LASSO2	-0.040	0.867	-0.040	0.919	-0.039	0.923	-0.038	0.914	-0.039	0.917
N=500	ENET1	-0.040	0.925	-0.041	0.925	-0.041	0.930	-0.040	0.923	-0.040	0.924
N=500	ENET2	-0.030	0.924	-0.031	0.927	-0.031	0.930	-0.030	0.924	-0.030	0.924

A tibble: 7 x 7

method `N=250, rho=0` `N=250, rho=0.35` `N=250, rho=0.7` `N=500, rho=0`

<chr> <chr> <chr> <chr> <chr>

## 1 P-Value	0 (0.945)	0.001 (0.939)	0.002 (0.935)	0 (0.949)
## 2 AIC	0 (0.944)	0 (0.938)	0.002 (0.935)	0 (0.948)
## 3 BIC	0 (0.979)	0.001 (0.975)	0.003 (0.973)	0 (0.987)
## 4 LASS01	0 (0.947)	0.011 (0.955)	0.017 (0.965)	0 (0.949)
## 5 LASS02	0 (0.972)	0.012 (0.987)	0.02 (0.992)	0 (0.982)
## 6 ENET1	0 (0.959)	0.016 (0.99)	0.021 (0.993)	0 (0.966)
## 7 ENET2	0 (0.947)	0.01 (0.951)	0.016 (0.96)	0 (0.949)
## # i 2 more variables:	`N=500, rho=0.35` <chr>, `N=500, rho=0.7` <chr>			

Conclusion

Summarize, next steps

Sources

<https://academic.oup.com/biometrics/article/80/3/ujae101/7758501?login=true>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3994246/>